

Electron Beam Polarimetry: Status and Prospects

DIS 2005, Madison, April 2005

E. Chudakov (JLab)

- **Motivation:** what accuracy is required for various experiments
- **Methods in use:**
 - Optical methods
 - Mott scattering
 - Moller scattering
 - Compton scattering
- **Most accurate polarimeters and ways to improve**
- **Conclusion**

Demand for Electron Beam Polarimetry and Accuracy Required

Double Spin Experiments

- $\vec{e}^- \vec{N} \rightarrow e^- X$ DIS - PDF, GDH
- $\vec{e}^- \vec{N} \rightarrow e^- N$ elastic: formfactors
- $\vec{e}^- N \rightarrow e^- \vec{N}$ elastic: formfactors
- $\vec{e}^- e^+ \rightarrow X$ SM

• $A_{obs} \propto \mathcal{P}_{beam} \times \mathcal{P}_{target} \times A_{reaction}$

Typically $\frac{\sigma(\mathcal{P})}{\mathcal{P}}_{target} > \frac{\sigma(\mathcal{P})}{\mathcal{P}}_{beam}$

Required: $\frac{\sigma(\mathcal{P})}{\mathcal{P}}_{beam} < 2 - 3\%$

Single Spin Experiments

- Neutral currents - parity violation (PV)
 - $\vec{e}^- N \rightarrow e^- X$ DIS - SM tests
 - $\vec{e}^- p \rightarrow e^- p$ formfactors, SM
 - $\vec{e}^- A \rightarrow e^- A$ nuclear physics
 - $\vec{e}^- e^+ \rightarrow X$ SM
- Charged currents $\vec{e}^- p \rightarrow \nu X$ - SM

• $A_{obs} \propto \mathcal{P}_{beam} \times A_{reaction}$

Required: $\frac{\sigma(\mathcal{P})}{\mathcal{P}}_{beam} < 1\%$

Electron Beam Polarimetry: Status and Prospects

Challenges to Polarimetry at JLab

Parity violating electron scattering experiments:

- Progress in reducing the systematic errors
- Beam polarization: becomes the dominant error!

JLab planned experiments, beam current $50\text{-}100\mu\text{A}$:

Experiment	Syst. err without pol	Polar. error	Stat. error	Energy GeV	Comments
^{208}Pb n-skin	0.5%	1.0%	3.0%	0.85	soon
eP $\sin^2\theta_W$	2.4%	<1.4%	2.8%	1.16	soon
DIS $\sin^2\theta_W$	0.3%	<1.0%	0.8%	10.0	after 12 GeV upgrade

A **1%**, polarimetry accuracy is needed A **0.5%** accuracy would be even better...

Electron Beam Polarimetry: Status and Prospects

Important Features of Beam Polarimetry

- Energy range $E_{beam} = 0.5 - 50 \text{ GeV}$, in future ILC $E_{beam} > 250 \text{ GeV}$
 - Low energy range $E_{beam} = 0.1 - 5 \text{ MeV}$ needed to test the injectors
 - Current range $E_{beam} = 0.01 \mu A - 2 \text{ A}$, various duty cycles
-
- Systematic error
 - Statistical error for a period of a possible polarization change
 - Invasive or not
 - Does polarimetry use the same beam (energy, current, location) as the experiment?

Methods Used for Absolute Electron Polarimetry

Spin-dependent processes with a known analyzing power.

Atomic Absorption

$\vec{e}^- \sim 50 \text{ keV}$ decelerated to $\sim 13 \text{ eV}$ $\vec{e}^-(13\text{eV}) + Ar \rightarrow A\vec{r}^* + e^-, A\vec{r}^* \rightarrow Ar + (h\nu)_\sigma$

Atomic levels: $(3p^5 4p)^3 D_3 \rightarrow (3p^6 4s)^3 P_2$ 811.5nm fluorescence

Potential $\sigma_{\text{sys}} \sim 1\%$. Under development (Mainz) - only relative so far.

Currently - invasive, diff. beam

Spin-Orbital Interaction

Mott scattering, 0.1-10 MeV: $e^- \uparrow + Z \rightarrow e^- + Z$ $\sigma_{\text{sys}} \sim 3\%$, $\Rightarrow 1\%$ (?)

invasive, diff. beam

Spin-Spin Interaction

- Møller scattering: $\vec{e}^- + \vec{e}^- \rightarrow e^- + e^-$ at $>0.1 \text{ GeV}$, $\sigma_{\text{sys}} \sim 3-4\%$, $\Rightarrow 0.5\%$

Mostly invasive, diff. beam

- Compton scattering: $\vec{e}^- + (h\nu)_\sigma \rightarrow e^- + \gamma$ at $>0.5 \text{ GeV} \sim 1-2\%$, $\Rightarrow 0.5\%$.

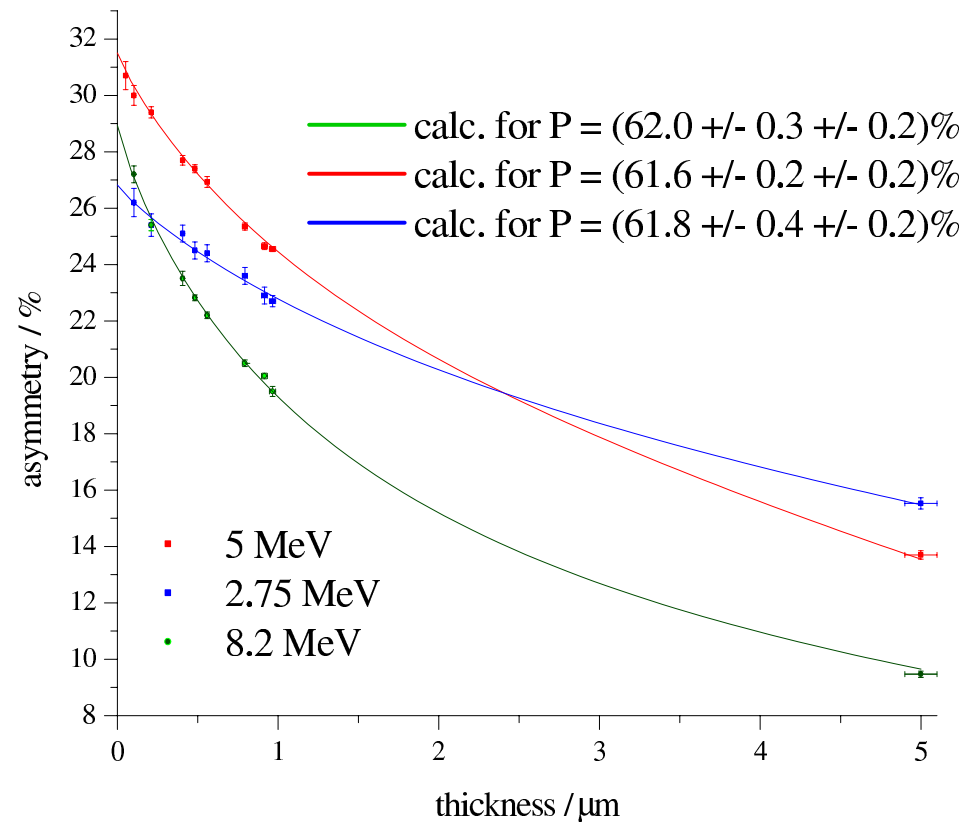
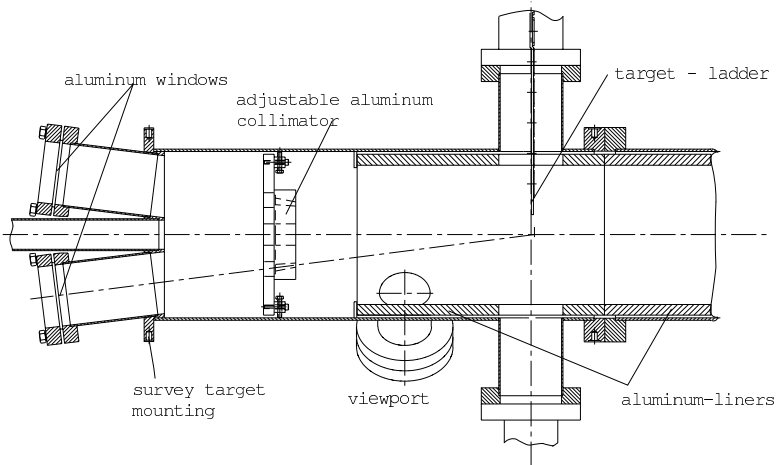
non-invasive, same beam

Electron Beam Polarimetry: Status and Prospects

Mott Polarimetry

0.1-10 MeV: $e^- \uparrow + Au \rightarrow e^- + Au$ Analyzing power - Sherman functions $\sim 1-3\%$:

- Nucleus thickness: phase shifts of scat. amplitudes
- Spin rotation functions
- Electron screening, rad. corrections
- Multiple and plural scattering
- No energy loss should be allowed
- Single arm - background



- Extrapolation to zero target thickness
- $e^- \uparrow < 5 \mu A$ - extrapolation needed

JLab: $\sigma(P)/P = 1\%(Sherman) \oplus 0.5\%(other)$ (unpublished) $\oplus \sigma(extrapol)$

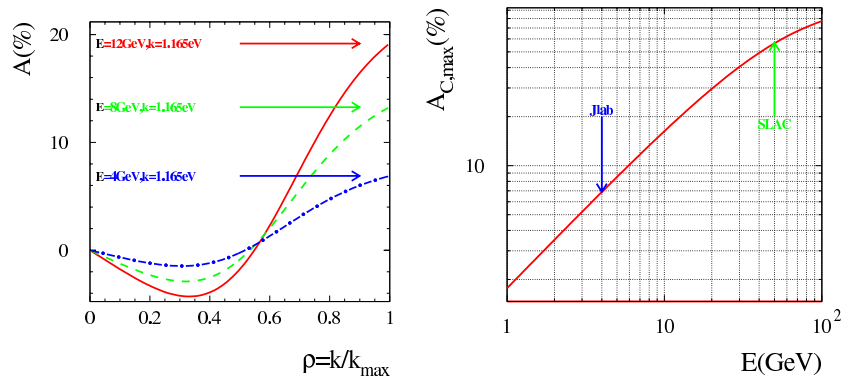
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Compton Polarimetry

$$\frac{\sigma_{\uparrow\uparrow} - \sigma_{\uparrow\downarrow}}{\sigma_{\uparrow\uparrow} + \sigma_{\uparrow\downarrow}} = A \cdot \mathcal{P}_b \mathcal{P}_t$$

Møller Polarimetry

$$\vec{e}^- + (h\nu)_\sigma \rightarrow e^- + \gamma \text{ QED.}$$



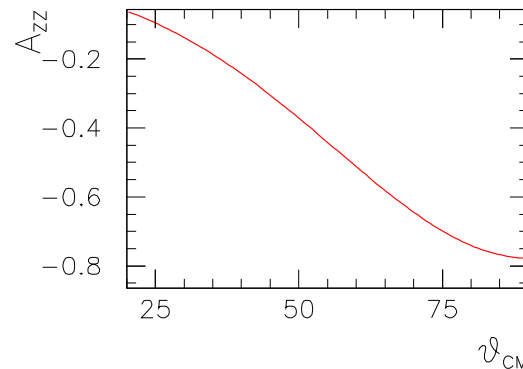
- Rad. corrections to Born $< 0.1\%$
- Detecting the γ at 0 angle
- Detecting the e^- with an energy loss
- Strong $\frac{dA}{dk'}$ - good $\sigma E_\gamma / E_\gamma$ needed
- $A \propto kE$ at $E < 20 \text{ GeV}$
- $T \propto 1/(\sigma \cdot A^2) \propto 1/k^2 \times 1/E^2$
- $\mathcal{P}_{laser} \sim 100\%$
- Non-invasive measurement

Syst. error 3→50 GeV: $\sim 1. \rightarrow 0.5\%$

Hard at $< 1 \text{ GeV}$: (JLab project) $\sim 0.8\%$

DIS 2005, Madison, April 29, 2005

$$\vec{e}^- + \vec{e}^- \rightarrow e^- + e^- \text{ QED.}$$



$$A(E) = -\frac{7}{9}$$

$$\sigma_{lab} \sim 180 \frac{mb}{ster}$$

- Rad. corrections to Born $< 0.3\%$
- Detecting the e^- at $\theta_{CM} \sim 90^\circ$
- $\left. \frac{dA}{d\theta_{CM}} \right|_{90^\circ} \sim 0$ - good systematics
- Beam energy independent
- Coincidence - no background
- Ferromagnetic target $\mathcal{P}_T \sim 8\%$
 - $> 1 \mu m$: invasive measurement
 - Beam $I_B < 2 - 4 \mu A$ (heating)
 - Levchuk effect
 - Low $\mathcal{P}_T \Rightarrow$ dead time
 - Syst. error $\sigma(\mathcal{P}_T) \sim 3\%$ (0.3%)

Syst. error $\sim 3\%$ typically, (0.5%)

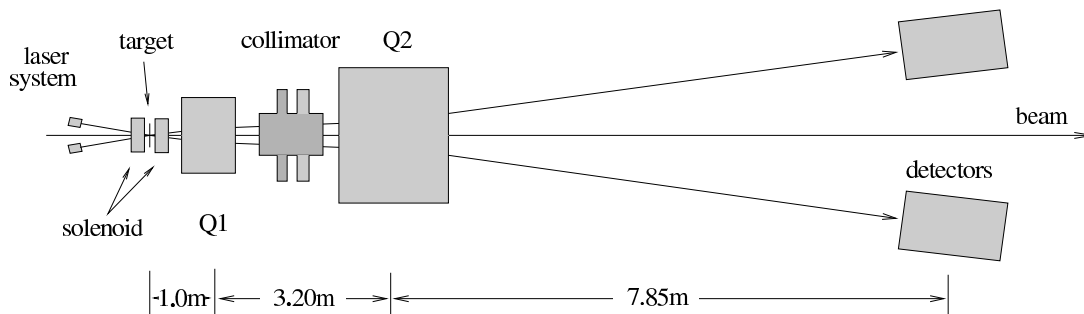
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Electron Beam Polarimetry: Status and Prospects

Møller Polarimeter with Saturated Iron foil

JLab, Hall C, M. Hauger *et al.*, NIM A **462**, 382 (2001)

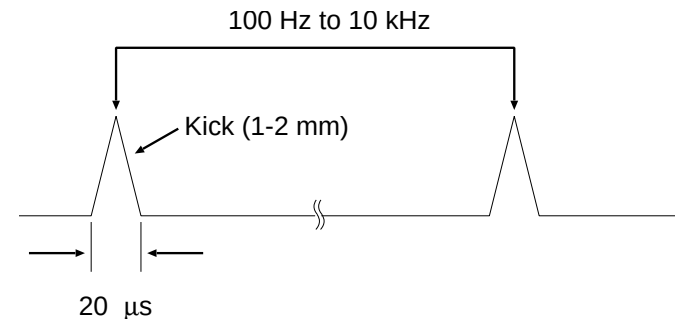
- External $B_Z \sim 4\text{ T}$
- Target foils 4-10 μm , perp. to beam
- $\mathcal{P}_t$ not measured
- Important: annealing, etc.



source	$\sigma(A)/A$
optics, geometry	0.20%
target	0.28%
Levchuk effect	0.30%
total	0.46%
$\Rightarrow 100\text{ }\mu\text{A}$	?

Tests for high current

- Beam $\sigma_X \approx 50\mu\text{m} > r = 12\mu\text{m}$
- At $20\mu\text{A}$ - accidentals/real ≈ 0.4
- $\sigma_{stat} \sim 1\%$ in 2h



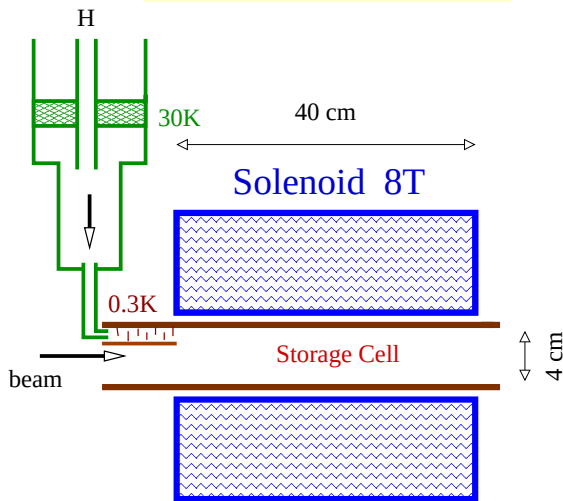
Current Studies

- A $1\mu\text{A}$ thick half-foil
- Higher duty factor

Møller Polarimeter with Atomic Hydrogen Target

Project: E. Chudakov and V. Luppov, IEEE Trans. Nucl. Sci. 51, 1533 (2004).

Ultra-cold traps



Atom H_1 : $\vec{\mu} \approx \vec{\mu}_e$, $E = -\vec{\mu}\vec{B}$

Population $\propto \exp(-E/kT)$

At 300 mK $\mathcal{P}_e \sim 1 - 10^{-5}$

Density $\sim 3 \cdot 10^{15} \text{ cm}^{-3}$

Lifetime $> 1 \text{ h}$

Stat. 1% in 10 min at 100 μA

Contamination and Depolarization at 100 μA CEBAF

- Hydrogen molecules $< 2 \cdot 10^{-5}$
- Upper states $|c\rangle$ and $|d\rangle < 10^{-5}$
- Excited states $< 10^{-5}$
- Helium, residual gas $< 0.1\%$ - beam-measurable
- Depolarization by beam RF $< 5 \cdot 10^{-5}$
- Ion, electron contamination $< 10^{-5}$
- Ionization heating $< 10^{-10}$

Expected depolarization $< 10^{-4}$

Limitations

- Problems $\propto \mathcal{I}_b^2/\mathcal{F} \Rightarrow$ “continuous” beam (MAMI, CEBAF ...)
- Complexity of the target

Advantages

- Expected accuracy $< 0.5\%$
- Non-invasive, continuous, the same beam

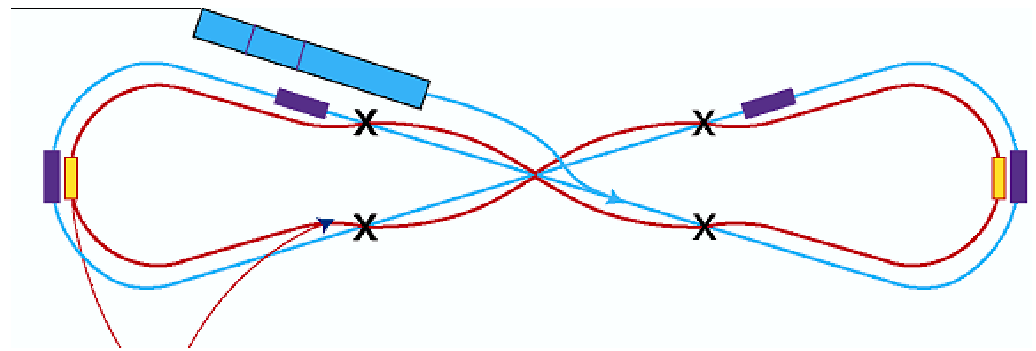
Bhabha/Møller Scattering at Colliders

Linear Collider

- Depolarization in bunch collisions $\Rightarrow$ measurement at the collision point!
- $\vec{e}^- \vec{e}^+$ 250 $\times$ 250 GeV
 $\mathcal{L} \sim 3 \cdot 10^{34} \text{ cm}^{-2}$
- Experiment $1 \cdot 10^7 \text{ s}$, $\mathcal{L} \sim 3 \cdot 10^5 \text{ pb}^{-1}$
- $\frac{d\sigma}{d\Omega_{CM} Bhabha} \sim 0.2 \text{ pb/Ster}$
- $\sim 10^5$ events $\Rightarrow \sigma(\mathcal{P})/\mathcal{P} \sim 0.7\%$
- W - production
- s-channel vector exchange - Blondel scheme

Faster methods are needed (Compton)

Electron-Ion Collider - JLab Design



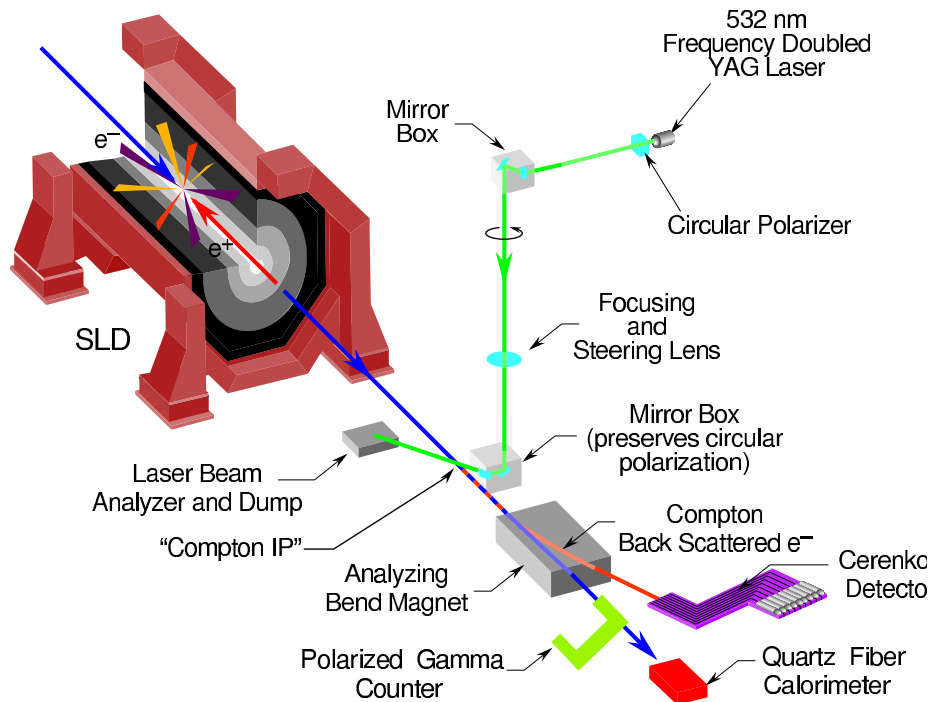
- $\vec{e}^- \vec{e}^-$ 3 $\times$ 3 GeV - 7 $\times$ 7 GeV
a new collision point at the center
- $\mathcal{L} \sim 1 \cdot 10^{35} \text{ cm}^{-2}$
- $\frac{d\sigma}{d\Omega_{CM} Møller} \sim 1000 \text{ pb/Ster}$
- Stat. 1% in 5 min
- A high luminosity collision point - some other interest?

Electron Beam Polarimetry: Status and Prospects

Compton Polarimeters: Best Accuracy at High Energy

SLAC SLD

Stat: 1% in 3 min



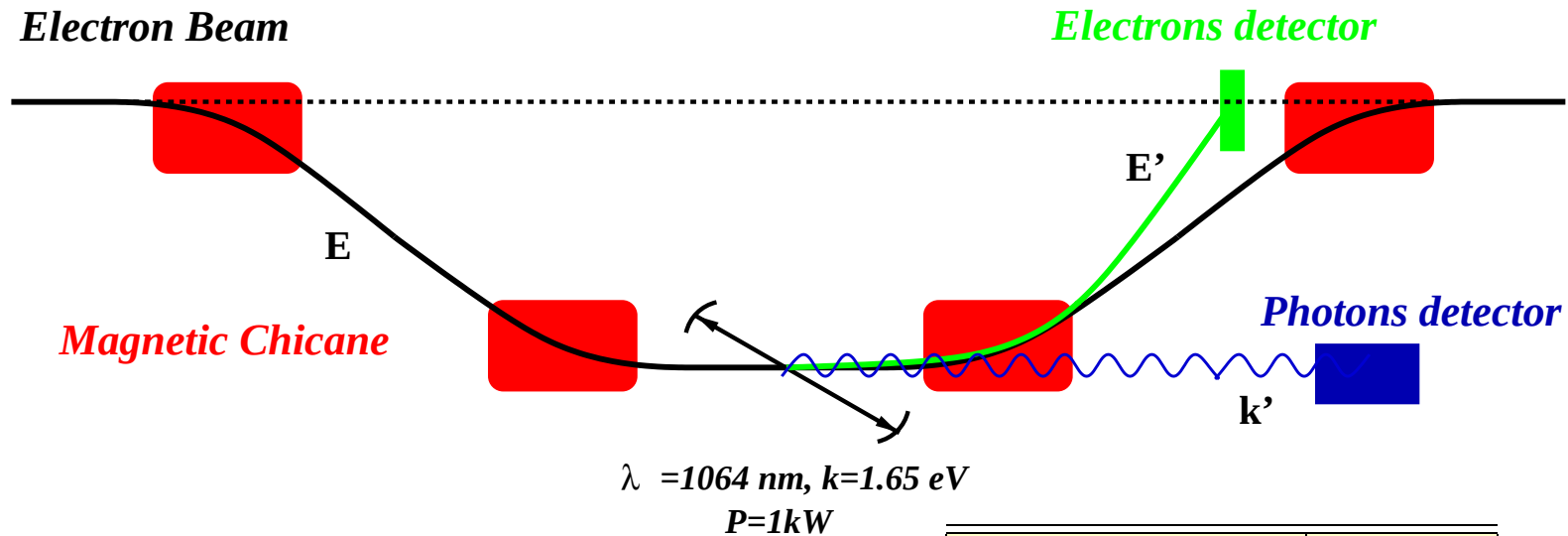
source	$\sigma(\mathcal{P})/\mathcal{P}$	
	SLD 1998	ILC Goal
Laser polarization	0.10%	0.10%
Analyzing power	0.40%	0.20%
Linearity	0.20%	0.10%
Electronic noise	0.20%	0.05%
total	0.50%	0.25%

M.Woods, JLab Polarimetry workshop, 2003

- Beam: 45.6 GeV
- Beam: $3.5 \cdot 10^{10} e^- \times 120 \text{ Hz} \sim 0.7 \mu\text{A}$
- Laser: 532 nm, 50 mJ at 7 ns $\times$ 17 Hz
- Crossing angle 10 mrad
- e^- 17-30 GeV detector - gas Cherenkov
- γ detector - calorimeter

Electron Beam Polarimetry: Status and Prospects

Compton Polarimeters: CW Low Energy Beam - Cavity



JLab Hall A Compton

- Beam: 1.5-6 GeV
- Beam: 5 – 100 μA at 500 MHz
- Laser: 1064 nm, 0.24 W
- Fabry-Pérot cavity $\times 4000 \Rightarrow 1 \text{ kW}$
- Crossing angle 23 mrad
- e^- detector - Silicon μ -strip
- γ detector - calorimeter

Stat: 1.0% in 30 min at 4.5 GeV, 40 μA

Syst: 1.2% at 4.5 GeV

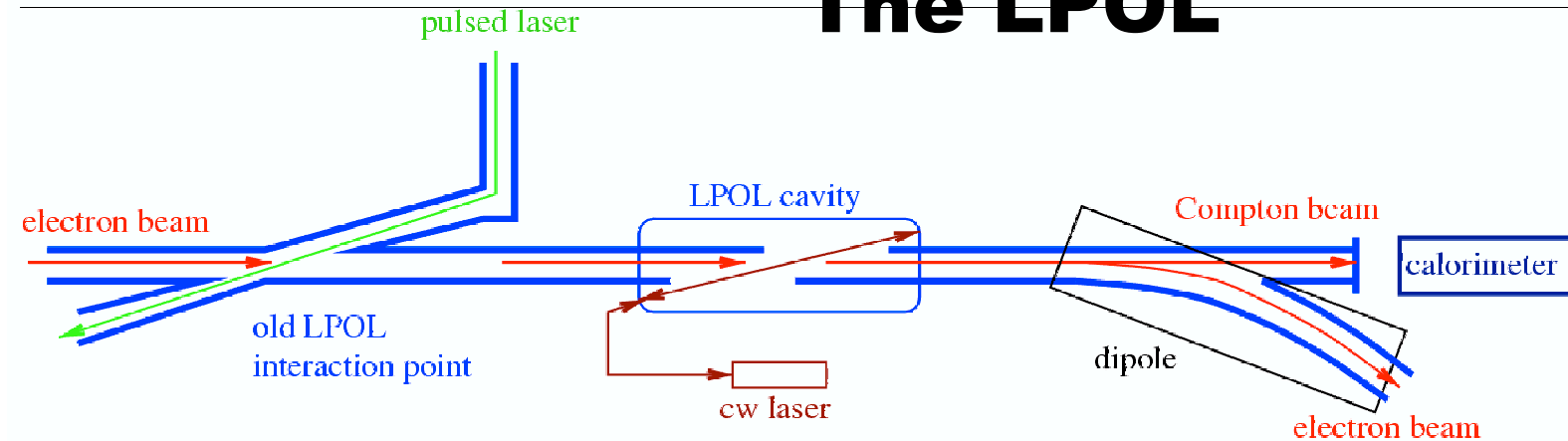
source	$\sigma(\mathcal{P})/\mathcal{P}$
Laser polarization	0.50%
Response function	0.40%
Calibration	0.60%
Others	0.65%
total	1.15%

Upgrade Plans - 1% at 0.85 GeV

- Laser: 532 nm, 0.1 W
- Cavity $\times 15000 \Rightarrow 1.5 \text{ kW}$
- Detector upgrade

Compton Polarimeters: Storage Rings (DESY)

The LPOL



Current

- $\sim 30 \text{ GeV } 3 \cdot 10^{10} e^- \times 10 \text{ MHz } \sim 40 \text{ mA}$
- Laser: 532 nm , 100 mJ at $< 100 \text{ Hz}$
- Crossing angle 8.7 mrad
- γ detector - calorimeter

Stat: 1.0% in 2 min in multi-photon mode

Syst: 1.6%

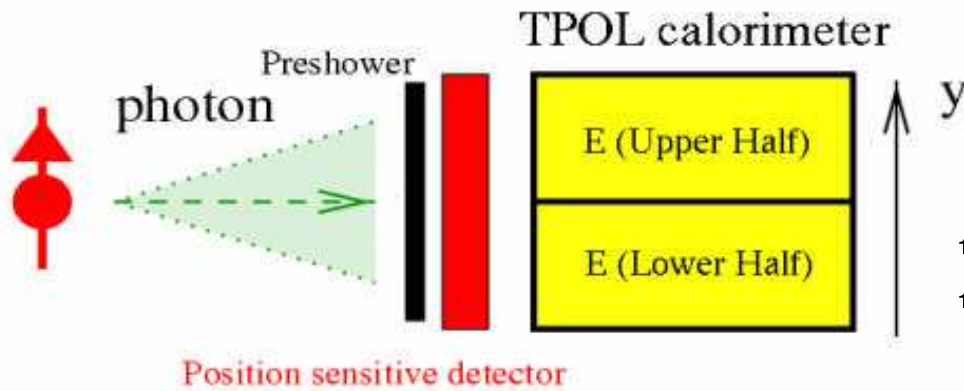
New Polarimeter

- Cavity 1064 nm (as JLab)
- Single bunch measurement
- Expected systematics: $\sim 0.1\%$

Electron Beam Polarimetry: Status and Prospects

Compton Polarimeters: Storage Rings (DESY)

TPOL

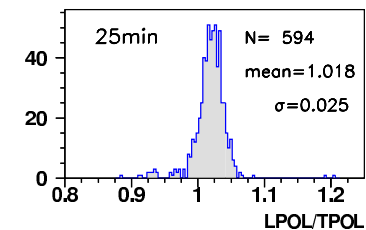
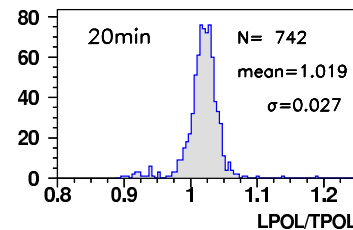
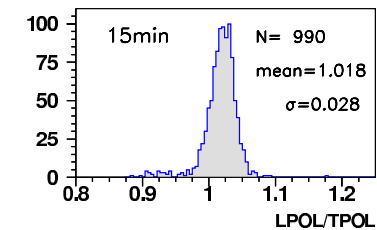
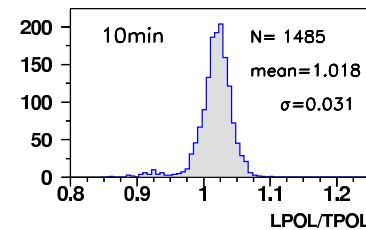
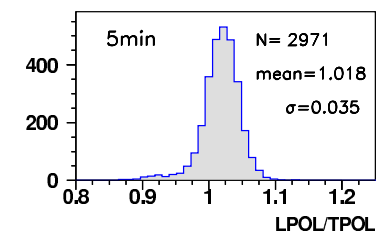
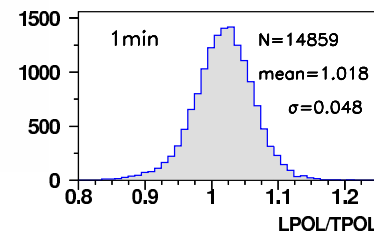


- Transverse polarization
- Laser: 532 nm CW
- Crossing angle 3.1 mrad
- Single photon mode
- γ detector - calorimeter

Syst: 1.9%

TPOL vs LPOL

- Stable for HERA-I
- Difference < 2%



Electron Beam Polarimetry: Status and Prospects

Conclusion

- Accuracy of $\sim 1.0\%$ becomes common for Compton polarimetry
- Accuracy of $\sim 0.5\%$ is still exceptional
- Compton polarimetry fits very well the high energy storage rings
- Moller polarimetry can be used at low energy, one may be able to reach $< 0.5\%$